Variance-Corrected Single-Index Composition for Heavy-Tailed Return Generators

Jeffrey D. Varner
Robert Frederick Smith School of Chemical and Biomolecular Engineering
Cornell University, Ithaca, NY 14850
jdv27@cornell.edu

Abstract

Per-asset generators for synthetic financial returns, such as the hidden Markov models with explicit jump components used in the `JumpHMM.jl` package, are built to preserve marginal structure (heavy tails, regime switching) but say nothing about cross-sectional factor dependence. The standard fix is to compose each generator draw with a single-index market factor, but the naive recipe inflates the marginal variance by a term that grows quadratically in the factor loading, distorting the very marginal the generator was designed to preserve; replacing the generator residual with a Gaussian draw eliminates the distortion but discards the marginal entirely. We derived a closed-form variance correction that sits between these two failure modes, rescaling the idiosyncratic draw by a scalar chosen so that the composed series inherits the generator's marginal variance exactly while recovering the calibrated factor loading. The correction was paired with a sign-preserving clip that engaged whenever the market component would have exceeded the generator variance, keeping defensive assets defensive, and a separate branch for index-tracking assets targeted the calibrated coefficient of determination rather than the marginal, with the SPY-on-itself identity recovered as a degenerate limit without special-casing. Branch selection was driven entirely by the calibrated coefficient of determination, requiring no external metadata, and each asset carried a persisted construction flag so downstream consumers could audit which path it took. Evaluated on a 424-ticker US equity and exchange-traded fund (ETF) universe against the SPY market path with `JumpHMM.jl` marginals, the hybrid construction recovered the factor loading to a median absolute error of 0.018 and passed a per-ticker Kolmogorov–Smirnov test against the real marginal at rate 50%, versus 4% for the naive composition and below 1% for the Gaussian-residual baseline. Three alternative generators fit directly to OLS residuals (JumpHMM-on-residuals, Politis–Romano block bootstrap, and GARCH(1,1)-t) achieved higher KS pass rates of 67–76%; on a per-ticker one-day 99% Value-at-Risk backtest, hybrid and residual-fit JumpHMM were statistically indistinguishable (mean exceedance 0.95% and 0.99% against the nominal 1%, Kupiec unconditional-coverage pass rates 80% and 83%). Hybrid is therefore the appropriate choice when the per-asset generator is specified in full-return units (as generators often are, for distributional interpretability or calibration reuse); the residual-fit alternatives are stronger on marginal body fidelity when the workflow permits a dedicated residual generator. All artifacts, scripts, and the fitted marginal cache are released with the paper.

Keywords: single-index model; synthetic return generation; hidden Markov models; heavy-tailed marginals; variance correction; factor composition.

Preprint.

1 Introduction

Portfolio construction, risk attribution, and stress testing often take long synthetic time series of asset growth rates as inputs. A good generator has to satisfy two things the downstream consumer cares about. First, each single asset's synthetic series should look like its real counterpart when you histogram it day by day: the tails should be as heavy as real tails, the kurtosis should match, regimes and jumps should show up with roughly the right frequency; in statistical language this is the *marginal distribution* of the asset. Second, if we stack synthetic series across many assets they should co-move the way real markets co-move: on a day when the market plunges, stocks with high factor loadings should plunge with it, tail co-movement should look realistic, and factor structure should be intact; this is the *cross-sectional dependence* across assets. These two requirements are largely orthogonal, and off-the-shelf generators tend to handle only one of them. Per-asset generators built from hidden Markov models [1], generalized autoregressive conditional heteroskedasticity (GARCH) families [2], or empirical bootstraps [3] handle the first: fit one of these models to AAPL's growth-rate history and its synthetic paths carry AAPL's tail thickness, kurtosis, and regime structure. But the fit knows nothing about market factors, e.g., SPY, or any other assets, e.g., MSFT. Stacking the per-asset draws therefore produces one synthetic growth-rate path per asset, each matching its own asset's full growth-rate marginal. But the paths carry no joint structure: on a day when synthetic SPY plunges, synthetic AAPL has no reason to plunge with it (but real AAPL would).

The standard recipe for adding factor structure is a linear factor model, either the one-factor single-index model (SIM) of Sharpe [4] or a multi-factor variant such as Ross's arbitrage pricing theory [5] or Fama-French [6]. We develop the construction for the one-factor SIM, which writes each asset's growth rate as an intercept plus a market-loaded term plus an idiosyncratic residual; the variance correction we propose extends to the multi-factor case without structural change. Applied naively, the recipe plugs a synthetic full growth-rate path from the per-asset generator (fit to the asset's real growth-rate history and matching its full growth-rate marginal) straight into the SIM as the idiosyncratic residual. Because the full growth rate already contains the market-driven part, using it as the residual double-counts the market variance and inflates the marginal variance of the composed series by an amount that grows quadratically in the factor loading. For high-loading assets the inflation is large enough to be visible in the empirical marginal: the body widens, the kurtosis drops, and the heavy tails that the generator was built to deliver are pushed out into a Gaussian-flavored distribution driven by the market term, defeating the purpose of fitting a heavy-tailed generator in the first place. The dual failure mode, drawing the residual from a Gaussian with variance taken from the real ordinary least squares (OLS) fit, recovers the SIM coefficients exactly but erases the marginal structure entirely. Neither naive option satisfies both targets at once, and closing that gap is the problem the rest of this paper addresses.

This paper describes a closed-form variance correction that sits between the two failure modes. The construction rescales the per-asset generator draw by a scalar chosen so that the variance algebra works out to preserve the generator marginal; the scalar admits a closed form whenever the market component accounts for less than the full generator variance. When that condition fails (high-loading assets paired with a low-volatility generator, or a synthetic market more volatile than the calibration market), the factor loading is clipped downward against a floor that reserves a configurable share of the marginal variance for idiosyncratic noise, with a sign-preserving clip that keeps defensive assets defensive. A separate branch handles index-tracking assets, where the calibrated coefficient of determination (the R^2 from OLS of the asset's real growth-rate history against the market path) rather than the generator marginal is the quantity worth preserving; SPY against itself recovers as the degenerate perfect-tracking limit with no special-casing. Branch selection is driven entirely by the calibrated R^2, and each asset carries a persisted construction flag (`hybrid`, `hybrid-clipped`, `r2-preserve`) so downstream consumers can audit which path each asset took. The procedure generalizes to any new ticker without external metadata.

The distinction between this paper and the companion `JumpHMM.jl` validation paper [7] deserves explicit statement. The companion paper fits and validates the per-asset JumpHMM marginals themselves: Student-t emissions, jump parameters, Laplace-partition state counts, per-ticker tuning, and cross-sectional copula specifications on the stacked outputs. Its question is whether the generator reproduces real marginal and joint statistics when used as-is. The present paper takes the generator as given and asks a separate question: once a per-asset generator hits the marginal target in isolation, how should it be composed with a single-index market factor so that neither the marginal nor the factor

relationship is broken? The variance correction, the sign-preserving clip rule, the R^2-preserving branch, the auditable construction flag, and the empirical comparison of composition rules on a common universe are specific to the present work and do not appear in the companion. The two contributions are orthogonal: the companion builds the per-asset generator; this paper builds the composition rule that lets that generator enter a factor-model pipeline without the marginal-versus-factor tradeoff.

In this study, we tested the construction on the 424-ticker US equity universe and SPY market path used by the companion `JumpHMM.jl` validation paper of Alswaidan and Varner [7], comparing six composition recipes on the same generators and the same market path: the naive composition, a Gaussian-residual SIM that recovers the calibrated factor coefficients and R^2 but loses the marginal structure, the hybrid construction proposed here, a JumpHMM marginal fit directly to the OLS residual series (composed with no variance correction), a Politis–Romano stationary bootstrap of the real residual history [3], and a per-ticker GARCH(1,1)-t fit to residuals [2]. We measured SIM recovery, marginal fidelity, and per-ticker one-day Value-at-Risk accuracy, reporting each per construction-flag branch. All six composers recovered the factor loading to a median absolute error of 0.018 (hybrid, residual-fit, bootstrap, GARCH) or 0.022 (naive). On marginal fidelity the hybrid construction held the Kolmogorov–Smirnov pass rate at 50%, against 4% for the naive composition and below 1% for the Gaussian baseline; the three residual-fit alternatives attained pass rates of 67 to 76%. On the 99% Value-at-Risk backtest, hybrid and JumpHMM-on-residuals delivered statistically indistinguishable coverage (Kupiec pass rates 80% and 83% respectively); the strawman composers failed coverage at around 45%. The broader implication is that the marginal-versus-factor tradeoff long assumed in SIM composition is not fundamental: a single closed-form scalar correction closes it to within a few percentage points of a dedicated residual-fit generator (JumpHMM-on-residuals, block bootstrap, or GARCH(1,1)-t) on marginal-body fidelity, and matches the residual-fit generator on a one-day 99% Value-at-Risk backtest against the real growth-rate histories. The hybrid composition rule trades a small body-fidelity gap for generator agnosticism: any per-asset full-return marginal, including heavy-tailed HMM-with-jumps or regime-switching generators calibrated once for other purposes, can enter the factor pipeline unchanged. The correction generalizes to richer factor structures (multi-factor SIM, lagged-factor SIM) without changing the underlying algebra.

2 Method

We work throughout in continuously compounded annualized growth rates, $g(t) = [\log p(t) - \log p(t-1)]/\Delta t$, where $p(t)$ is the closing price on trading day t and Δt is the fraction of a trading year spanned by a single trading day (we use $\Delta t = 1/252$ in the experiments below, so the per-day log return is $g(t)\,\Delta t$). Working in annualized growth rates rather than raw log returns keeps variances and covariances in the units customary in risk reporting. Growth rates compose linearly under aggregation in time and are the natural unit for variance and covariance arithmetic, which the construction below relies on heavily. All variances and covariances refer to sample moments computed over the relevant time window unless stated otherwise. The method composes a synthetic asset growth-rate path of the form:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{1}$$

from an independent generator draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ and a shared market path $g_m \in \mathbb{R}^T$ over T trading days by rescaling the generator draw as $\varepsilon_i = s_i\,\tilde{\varepsilon}_i$ with a scalar $s_i \in (0, 1]$ chosen to preserve one of two per-asset marginal targets while recovering the calibrated single-index model (SIM) coefficients (α_i, β_i) in the large-T limit.

2.1 Problem setup

We are given a single market growth-rate path $g_m \in \mathbb{R}^T$ over T trading days, with sample mean μ_m and sample variance $\sigma_m^2 = \mathrm{Var}[g_m]$. In the experiments of Section 3 the market path is the SPY total-return growth-rate series; the construction is agnostic to whether g_m is a single-name proxy, a synthetic index, or a model-generated path.

For each of N assets indexed by $i = 1, \ldots, N$, we are given a draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ from a generator that produces growth rates with the target marginal distribution: heavy tails, regime structure, jumps, or whatever the application requires. Write $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}[\tilde{\varepsilon}_i]$ for the sample variance of the draw. The construction is agnostic to how the generator is fit; in our experiments we use the per-asset

hidden Markov model with Student-t emissions and explicit jump components from `JumpHMM.jl`[1], fit to the asset's own full growth-rate history (not to residuals after market detrending). Because the generator's variance reflects the full return rather than the idiosyncratic part, the rescaling $\varepsilon_i = s_i\,\tilde{\varepsilon}_i$ with $s_i^2 = 1 - \beta_i^2\sigma_m^2/\sigma_{\mathrm{gen},i}^2$ is exactly what prevents the market term in the SIM composition from double-counting variance that is already carried by $\tilde{\varepsilon}_i$. The generator draws are demeaned, $\mathbb{E}[\tilde{\varepsilon}_i] = 0$, so that the constant term in the SIM is carried entirely by α_i. We assume $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$ for each i, which holds when the generator and market path are independent draws (the typical use case), and holds after a copula rank reorder when the copula is sampled independently of g_m (verified empirically across the 423-ticker universe in Fig. S1). If the generator already carries non-trivial market correlation, the residual covariance can be subtracted from β_i before applying the correction; we return to this in Section 4.

For each asset we are given a calibrated triple $(\alpha_i, \beta_i, R^2_{i,\mathrm{real}})$, estimated by ordinary least squares on the real return history of asset i against the real market path. These are the targets we want the synthetic construction to recover. The intercept α_i and slope β_i are the standard SIM coefficients; $R^2_{i,\mathrm{real}}$ is the real-data coefficient of determination, used by the branch-selection step in Section 2.5.

2.2 Preservation criteria

We want the synthetic composition $g_i = \alpha_i + \beta_i^{\mathrm{eff}}\,g_m + \varepsilon_i$, with effective slope β_i^{eff} (equal to β_i on the unclipped branch and the attenuated value of Eq. (6) otherwise), to satisfy three properties in the large-T limit. The first is *SIM recovery*: ordinary least squares of g_i on g_m returns the calibrated intercept and slope, $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\mathrm{eff}}$. On the unclipped branch, $\beta_i^{\mathrm{eff}} = \beta_i$ and the OLS output reproduces the calibration exactly; on the clipped branch, β_i^{eff} is the attenuated value defined by Eq. (6), and the attenuation is persisted alongside the output so it is auditable. The second is *marginal fidelity*: each asset is assigned one of two per-asset targets based on its calibrated R^2, not both at once. Assets whose generator marginal (heavy tails, regime structure, jumps) is the quantity of interest receive the variance-preserving target $\mathrm{Var}[g_i] = \sigma^2_{\mathrm{gen},i}$; assets that track the market closely receive the R^2-preserving target $R^2 = R^2_{i,\mathrm{real}}$, measured by the same OLS regression as above. The branch chosen for each asset is recorded as a construction flag so downstream consumers can audit which target was pursued. The third is *cross-sectional dependence*: any joint dependence structure applied to the residuals $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to the composed series $\{g_i\}_{i=1}^N$, automatic given the linearity of Eq. (1) in ε_i. The joint structure itself (empirical copula, factor copula, t-copula) is supplied as a separate design choice, and we report cross-sectional fidelity for the empirical-copula choice in Section 3.

The naive composition $g_i = \alpha_i + \beta_i\,g_m + \tilde{\varepsilon}_i$ satisfies SIM recovery and cross-sectional dependence but violates marginal fidelity: its marginal variance is $\sigma^2_{\mathrm{gen},i} + \beta_i^2\sigma_m^2$, an inflation that grows quadratically in β_i. The rest of this section fixes this inflation with a closed-form scalar correction, handles the degenerate high-β case with a sign-preserving clip, treats index-tracking assets through an R^2-preserving branch, and ties the pieces together in an algorithm that also states the resulting properties of the composed series.

2.3 Variance correction

Form the candidate residual $\varepsilon_i = s_i\,\tilde{\varepsilon}_i$ for some scalar $s_i \in (0, 1]$, and require that the composed series have the same variance as the generator,

$$\mathrm{Var}[\beta_i\,g_m + \varepsilon_i] \;=\; \sigma^2_{\mathrm{gen},i}. \tag{2}$$

Under the independence assumption of Section 2.1, the cross term in (2) vanishes and

$$\beta_i^2\,\sigma_m^2 \;+\; s_i^2\,\sigma^2_{\mathrm{gen},i} \;=\; \sigma^2_{\mathrm{gen},i}, \tag{3}$$

giving

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\mathrm{gen},i}}} \tag{4}$$

[1] `https://github.com/varnerlab/JumpHMM.jl`

Define the *market loading ratio*

$$\rho_i \;\equiv\; \frac{\beta_i^2\,\sigma_m^2}{\sigma_{\mathrm{gen},i}^2}, \tag{5}$$

the share of the generator variance consumed by the market component. Then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$. The naive composition $g_i = \alpha_i + \beta_i g_m + \tilde{\varepsilon}_i$ corresponds to $s_i = 1$; it inflates $\mathrm{Var}[g_i]$ by exactly $\beta_i^2\sigma_m^2$, so the relative distortion grows quadratically in β_i.

2.4 Beta clipping with an idiosyncratic floor

Equation (4) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving no room for idiosyncratic noise. This case arises for high-β_i assets paired with low-volatility generators, or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated.

To handle it, impose a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clip β_i downward whenever the floor would be violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$,

$$\beta_i^{\mathrm{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \mathrm{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma_{\mathrm{gen},i}^2}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{6}$$

Sign preservation in (6) keeps defensive (negative-β) assets defensive even when clipped. The floor f is a tunable knob; in the experiments below we use $f = 0.10$, which guarantees that at least 10% of each asset's variance comes from idiosyncratic noise. When β_i is clipped we log a warning and persist both β_i and β_i^{eff} so downstream consumers can audit the attenuation.

2.5 R^2-preserving branch for tracker assets

The variance-preserving construction targets $\mathrm{Var}[g_i] = \sigma_{\mathrm{gen},i}^2$. That is the right target when the generator marginal carries information one wants to keep. For some assets the generator marginal is not really the quantity of interest; the SIM relationship to the market is. Index ETFs are the canonical example. SPY against itself has $\beta = 1$, $R^2 = 1$, $\sigma_\varepsilon = 0$ exactly. QQQ against SPY has $R^2 \approx 0.84$, SPYG against SPY has $R^2 \approx 0.86$. For these tickers the practitioner cares that $\hat{\beta}$ recovers to its calibrated value and that R^2 recovers to its calibrated value, neither inflated to one by dropping ε nor implied by unrelated marginal-variance bookkeeping.

We branch on the real-data calibration $R_{i,\mathrm{real}}^2$. Pick a threshold R_{preserve}^2 that separates index-tracking assets from genuine single stocks; in our universe the cumulative distribution sorts itself cleanly, with index ETFs above 0.80 and the most market-like single names below 0.65, so $R_{\mathrm{preserve}}^2 = 0.80$ captures the trackers without sweeping in any genuinely idiosyncratic name. For any asset with $R_{i,\mathrm{real}}^2 \geq R_{\mathrm{preserve}}^2$, the population R^2 identity for (1),

$$R^2 \;=\; \frac{\beta_i^2\,\sigma_m^2}{\beta_i^2\,\sigma_m^2 + \sigma_{\varepsilon,i}^2}, \tag{7}$$

inverts to give the residual variance that hits the calibrated R^2:

$$\boxed{\sigma_{\varepsilon,\mathrm{target}}^2 \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1 - R_{i,\mathrm{real}}^2}{R_{i,\mathrm{real}}^2}} \tag{8}$$

Draw $\tilde{\varepsilon}_i$ from the ticker's generator and rescale,

$$\varepsilon_i \;=\; \sqrt{\frac{\sigma_{\varepsilon,\mathrm{target}}^2}{\sigma_{\mathrm{gen},i}^2}}\,\tilde{\varepsilon}_i, \tag{9}$$

then copula-reorder and compose as in (1). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R_{i,\mathrm{real}}^2$ by construction.

Algorithm 1 Hybrid SIM composition with variance correction.

Require: Market path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator draws $\tilde{\varepsilon}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

1: **for** each asset $i = 1, \ldots, N$ **do**
2: Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \text{Var}[\tilde{\varepsilon}_i]$.
3: **if** $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
4: $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}}) / R^2_{i,\text{real}}$ {Eq. (8)}
5: $\varepsilon_i \leftarrow \sqrt{\sigma^2_{\varepsilon,\text{target}} / \sigma^2_{\text{gen},i}}\, \tilde{\varepsilon}_i$
6: $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag ← `r2-preserve`
7: **else**
8: $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ {Eq. (5)}
9: **if** $\rho_i \leq 1 - f$ **then**
10: $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag ← `hybrid`
11: **else**
12: $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i) \sqrt{(1 - f)\, \sigma^2_{\text{gen},i} / \sigma_m^2}$; flag ← `hybrid-clipped`
13: **end if**
14: $\varepsilon_i \leftarrow s_i\, \tilde{\varepsilon}_i$
15: **end if**
16: *(Optional)* apply cross-sectional rank reorder to ε_i.
17: $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}}\, g_m + \varepsilon_i$ {Eq. (1)}
18: **end for**

When $R^2_{i,\text{real}} = 1$ exactly (SPY against itself), (8) gives $\sigma^2_{\varepsilon,\text{target}} = 0$, so $\varepsilon_i \equiv 0$ and $g_i = \alpha_i + \beta_i g_m$ deterministically. No special-casing is needed: the SPY identity falls out of the same expression with $\beta = 1$, $\alpha = 0$, $R^2 = 1$. If the rescale factor in (9) exceeds one, the generator's tails are being stretched to fit a target larger than the original draw; this happens when the synthetic market g_m has lower variance than the calibration market. We detect and flag this case so the loss of marginal fidelity is auditable.

2.6 Branch selection, algorithm, and properties

The pieces combine into a single procedure (Algorithm 1). For each asset the calibrated $R^2_{i,\text{real}}$ selects a branch: assets with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ go through the R^2-preserving construction of Section 2.5, and the rest go through the variance-preserving construction of Section 2.3 with clipping (Section 2.4) engaged when the market loading ratio ρ_i would otherwise exceed $1 - f$. Each asset is tagged with a construction flag `r2-preserve`, `hybrid`, or `hybrid-clipped` that is persisted alongside the output, so downstream consumers can audit which path each asset took. The optional rank reorder in line 14 accepts any joint dependence structure (empirical copula, t-copula, factor copula); reordering preserves the marginal variance of ε_i, so the variance algebra of Section 2.3 is unaffected and whatever cross-sectional structure is applied to ε_i passes through to g_i unchanged.

By construction the composed series g_i satisfies the preservation criteria of Section 2.2 in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$, with $\beta_i^{\text{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\text{gen},i}$ on the variance-preserving branch, either exactly, $\text{Var}[g_i] = \beta_i^2 \sigma_m^2 + s_i^2 \sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, or by the choice of β_i^{eff} in (6) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2 \sigma_m^2 / R^2_{i,\text{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\text{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\, \sigma^2_{\text{gen},i}$ of the variance is reserved for ε_i, the tails inherited from $\tilde{\varepsilon}_i$ dominate whenever $(\beta_i^{\text{eff}})^2 \sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires

a richer construction, which we return to in Section 4. The pipeline is modest in compute: fitting the 423 per-asset JumpHMM marginals runs once offline against the 2014–2024 price histories; the scoring loop across $423 \times 100 = 42{,}300$ per-ticker replications per composer runs in minutes on a commodity laptop, and the full six-composer sweep reported in Section 3 completes within an hour on the same hardware.

3 Results

We evaluated the three composers on a fixed set of per-asset JumpHMM marginals calibrated against the real SPY growth-rate path. This isolated the effect of the composition rule from the effect of market-path randomness: all three composers consumed the same real SPY history and the same per-ticker innovation draw, so metric differences across composers reflected the composition rule alone. The ticker universe was a 424-asset US equity/ETF set drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024, the same window used by the companion `JumpHMM.jl` validation paper); filtering on the common length preserved 424 tickers, of which the SPY total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a `JumpHMM` marginal to the asset's own growth-rate history, using the default configuration ($N = 100$ Laplace-partition states, Student-t emissions with $\nu = 5$, annualized growth rates via `excess_growth_rates`, risk-free rate $r_f = 0$, $\Delta t = 1/252$); jump parameters (ϵ, λ) were left at the package defaults rather than tuned per ticker, which would improve per-ticker marginal fit but was not required for a comparison that tests the effect of the composition rule on top of whatever marginal the generator produces. For each non-market ticker we then ran ordinary least squares of the asset's growth-rate history on g_m, yielding the calibrated quadruple $(\alpha_i, \beta_i, R^2_{i,\mathrm{real}}, \sigma_{\varepsilon,\mathrm{real},i})$. The universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$; $R^2_{i,\mathrm{real}}$ ranged from 0.001 to 0.933 (Fig. 7), with only two tickers above the $R^2_{\mathrm{preserve}} = 0.80$ threshold (QQQ at $R^2 = 0.861$ and SPYG at $R^2 = 0.933$) and the highest-R^2 individual stock (BLK at $R^2 = 0.645$) a structural ~ 0.22 below; the branch assignment is therefore empirically tracker-specific on this universe, not a threshold artifact.

The composition rules used a *paired innovation* protocol where applicable: for each non-market ticker and each replication $r \in \{1, \ldots, R\}$ we drew a single innovation path $\tilde{\varepsilon}_i^{(r)}$ of length T from the ticker's JumpHMM marginal, and the same $\tilde{\varepsilon}_i^{(r)}$ was passed to the naive, Gaussian-residual-SIM, and hybrid composers, so that differences among those three reflect the composition rule and not generator stochasticity. The *Naive* composer set $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \tilde{\varepsilon}_i^{(r)}$ with no variance correction. The *Gaussian SIM* baseline replaced the JumpHMM draw with $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \sigma_{\varepsilon,\mathrm{real},i}\, \xi_i^{(r)}$, $\xi_i^{(r)} \sim \mathcal{N}(0, I_T)$ drawn fresh per replication, recovering the calibrated R^2 exactly but discarding any marginal structure carried by $\tilde{\varepsilon}$. The *Hybrid* composer applied Algorithm 1 with floor $f = 0.10$ and threshold $R^2_{\mathrm{preserve}} = 0.80$. We evaluated three further composers that draw their innovations from dedicated residual generators, each fit to the real OLS residual series $e_i(t) = G_i(t) - \alpha_i - \beta_i G_m(t)$. The *JumpHMM-on-residuals* composer fits a fresh JumpHMM marginal to e_i via a pseudo-price inversion (so the `JumpHMM.jl` [7] fit interface that expects a price series accepts the residual target without modification) and composes $g_i^{(r)} = \alpha_i + \beta_i g_m + \tilde{\varepsilon}^{(r)}_{\mathrm{resid},i}$ with no variance correction. The *Block bootstrap* composer draws a Politis–Romano stationary bootstrap replicate [3] of the empirical residual series with mean block length $\sqrt{T} \approx 50$ and composes analogously. The *GARCH(1,1)-t* composer fits a conditional-variance model per ticker [2] and substitutes a simulated residual path; 30 of the 423 tickers yielded a non-stationary GARCH fit ($\alpha + \beta \geq 1$ at the optimizer's termination) and were excluded from that composer only. All three composers skipped the optional cross-sectional rank reorder in Algorithm 1 so that per-ticker metrics reflected the marginal construction in isolation; cross-sectional dependence is studied separately in the companion `JumpHMM.jl` paper and is not the object of this evaluation. For every composed series we computed three metric families against the ticker's real growth-rate history and the real SPY path: SIM recovery reported $(\hat{\alpha}_i, \hat{\beta}_i, \hat{R}^2_i)$ from OLS of $g_i^{(r)}$ on g_m along with the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\mathrm{real}}|$; marginal fidelity was assessed by Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, Wasserstein-1 distance W_1, excess kurtosis κ, and the classical Hill tail-index

estimator $\hat{\xi} = (k-1)^{-1} \sum_{i=1}^{k-1} \log(X_{(i)}/X_{(k)})$ on the upper 5% order statistics of $|g_i^{(r)}|$, so reported Hill values are tail indices with $\hat{\xi} \approx 1/\alpha$ for a Pareto-tailed distribution with shape α; variance preservation was the ratio $\mathrm{Var}(g_i^{(r)})/\sigma^2_{\mathrm{gen},i}$, using the real-data variance as a proxy for the generator's targeted variance. Pass rates were reported at $\alpha = 0.05$, and we reported $R = 100$ replications per ticker so that aggregate metrics were averages over $423 \times 100 = 42{,}300$ per-ticker observations per composer; the simulation seed was fixed at 1234.

We began with the aggregate comparison across composers, computing per-asset medians of each metric and then aggregating across the universe (Table 1). Applying the protocol to the 423 non-market tickers yielded one composed series per $(\mathrm{ticker}, \mathrm{composer}, \mathrm{replication})$ triple, for a total of $423 \times 100 \times 3 = 126{,}900$ scored series. All three composers recovered β to within a median absolute error of 0.018 to 0.022, consistent with the identity $\mathrm{Cov}(g, g_m)/\sigma_m^2 \to \beta_i^{\mathrm{eff}}$ holding at all residual scales. The differences between composers appeared everywhere else. The Gaussian SIM baseline recovered R^2 most tightly (median error 0.008) because it targeted R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. 2). The naive composition retained the generator marginal but inflated its variance by $\beta_i^2\sigma_m^2$, so its KS and AD pass rates sat at 4.5% and 2.7%. The hybrid composer held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; the R^2 recovery (0.021) came in tighter than the naive baseline but looser than the Gaussian baseline, reflecting that the variance-preserving branch targeted $\sigma^2_{\mathrm{gen},i}$ rather than $R^2_{i,\mathrm{real}}$.

The three residual-fit composers sidestep the variance-inflation problem entirely by fitting a generator directly to e_i, which by construction has variance $\sigma^2_{\varepsilon,\mathrm{real},i}$ and no double-counted market component. Their aggregate KS pass rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4% (Table 1, Fig. 4). The advantage sat in the body of the distribution rather than the tails: hybrid's median excess kurtosis (7.17) and Hill upper-tail index (0.365) matched JumpHMM-on-residuals (7.26 and 0.365) to within sampling variability, confirming that the variance correction of Eq. (4) preserves the heavy-tailed character of the full-return generator even when the rescaling is applied. Block bootstrap and GARCH, by fitting their own residual generators, reproduced the empirical residual series but with slightly lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based recipe.

The downstream consequence of the body-fidelity gap was the question a per-ticker one-day Value-at-Risk backtest was designed to answer (Table 5, Fig. 9). At $\alpha = 0.99$ the nominal exceedance rate is 1%; the naive composer under-breached at 0.67% and the Gaussian baseline over-breached at 1.40%, both failing Kupiec unconditional coverage on the majority of the universe (45 and 47% pass rates). The hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t composers all hit coverage within one-tenth of a percentage point of nominal (0.95, 0.99, 1.09, and 1.09% respectively), and their Kupiec pass rates clustered between 72 and 86%. Hybrid and JumpHMM-on-residuals were statistically indistinguishable on this test: the 25pp body-fidelity gap between them did not translate to a Value-at-Risk accuracy difference. Hybrid's standalone advantage is therefore scoped: it is the right rule when a full-return per-asset generator is the given input, and it delivers tail accuracy on par with a dedicated residual generator without requiring a separate fitting stage.

To see where the KS advantage over the naive baseline came from, we next plotted per-ticker KS pass rate and per-ticker marginal variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ as functions of the calibrated β (Fig. 1). The variance-ratio panel made the mechanism explicit: the naive scatter traced the theoretical curve $1 + \rho_i$ with $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\mathrm{gen},i}$, overshooting the generator variance by 50% to 75% at high β, while hybrid and Gaussian SIM sat on the unit line by design. The KS-pass-rate panel showed the consequence: naive fell from a pass rate of ~ 0.25 at $\beta \approx 0.3$ to below 0.01 past $\beta \approx 0.7$, while hybrid sat between 0.3 and 0.9 across the full β range. Gaussian SIM was pinned to zero throughout; with its Gaussian residual it did not match the heavy-tailed real marginal at any β. Having traced the KS advantage to variance preservation, we next separated the two baselines' failure modes by plotting per-ticker excess kurtosis and Hill tail index against β (Fig. 2). The naive and hybrid composers clustered together near the real-data kurtosis reference line ($\kappa_{\mathrm{real}} \approx 13$), with the difference between them reflecting the variance inflation under naive rather than any difference in tail shape. The Gaussian SIM series collapsed to $\kappa \approx 1$ and Hill ≈ 0.22, pinned below the generator regardless of β; by replacing the heavy-tailed innovation with a Gaussian draw the baseline removed the entire heavy-tail carrier. The

Hill panel confirmed the same separation at the tail: naive and hybrid both reported ~ 0.37, while Gaussian SIM reported ~ 0.22.

Having established aggregate preservation, we next examined the branch-selection outcome on this universe and asked whether the advantage held across the β distribution. Applying the branch-selection rule of Section 2.6 to the 423-ticker universe produced 421 assignments to the `hybrid` branch, 2 to the `r2-preserve` branch, and 0 to the `hybrid-clipped` branch (Fig. 3, Table 2). The two R^2-preserving tickers were QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$), both ETFs that track market-capitalization indices closely; their KS pass rate under hybrid composition was 99%, consistent with the R^2-preserving branch recovering both the calibrated β and the calibrated R^2 by construction. Clipping did not trigger anywhere in the universe: the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stayed below $1 - f = 0.90$ for every ticker, because the JumpHMM marginals carried enough variance that the market component never crowded out the $f = 0.10$ idiosyncratic floor. Because the R^2-preserve branch had only two empirical assignments, we validated it on synthetic trackers where the target (β, R^2) was known by construction (Fig. 8). For each $(\beta_{\text{true}}, R^2_{\text{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$ we synthesized an asset via $g_i(t) = \alpha_{\text{true}} + \beta_{\text{true}} g_m(t) + \sigma_{\varepsilon,\text{true}}\, \xi(t)$ with $\sigma^2_{\varepsilon,\text{true}} = \beta^2_{\text{true}} \sigma_m^2 (1 - R^2_{\text{true}}) / R^2_{\text{true}}$ and ran the series through the hybrid composer at 100 replications. Across all 15 cells the branch selector assigned `r2-preserve` as intended, recovered median $\hat{\beta}$ to within 0.001 of β_{true} and median $\hat{R}^2$ to within 0.001 of R^2_{true}, and cleared the $\alpha = 0.05$ KS test on at least 99% of replications per cell. The branch is therefore correct by construction and by controlled test, with the two-ticker empirical occupancy reflecting the S&P 500 universe rather than a limitation of the composition rule. To check whether the preservation advantage held across the β distribution, we bucketed the universe by β quartile and recomputed the aggregate metrics within each bucket (Table 3). The hybrid KS pass rate declined gently from 67% in the low-β quartile to 44% in the high-β quartile; the Gaussian SIM pass rate stayed pinned near zero across all buckets; the naive pass rate collapsed from 16% in Q1 to under 1% in Q3 and Q4 as $\beta_i^2 \sigma_m^2$ grew. The gentle hybrid decline at high β reflected the caveat noted in Section 2.6: as $\beta^{\text{eff}\,2} \sigma_m^2$ grew toward $(1 - f)\sigma^2_{\text{gen}}$, the market-driven component of g tightened the marginal toward Gaussian along its own axis, mildly attenuating the generator's tails in the composed series. The effect was small enough that hybrid still beat both baselines by more than an order of magnitude in every bucket.

To probe the clipping branch, which did not trigger anywhere in the main run, we re-ran the hybrid composer on the same universe with the market growth-rate path rescaled by a factor $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals $(\alpha_i, \beta_i, \sigma_{\varepsilon,\text{real},i}, \sigma_{\text{gen},i})$ fixed and drawing $R = 100$ replications per (ticker, γ) pair (Fig. 5). The scale factor entered the market loading ratio as $\rho_i(\gamma) = \gamma^2 \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$, so increasing γ simulated an environment in which market variance grew without the per-ticker generators being refit. At $\gamma = 1$ no ticker crossed the clipping threshold and the hybrid KS pass rate matched the headline value (50.3%). Doubling σ_m pushed 76.4% of tickers past $\rho_i > 1 - f = 0.90$ and the clipping branch activated accordingly; at $\gamma = 3$ the clipping rate reached 95.2%. Hybrid KS pass rates under stress were 71.8% at $\gamma = 2$ and 77.3% at $\gamma = 3$, above the unclipped baseline, because the clipping branch rescales β^{eff} to hold the idiosyncratic floor $f\sigma^2_{\text{gen},i}$ and thereby preserves marginal body shape when the market component would otherwise crowd it out. The clipping branch is therefore not a degradation under stress but a mechanism that keeps the composed marginal near the generator target as $\beta^{\text{eff}\,2} \sigma_m^2$ inflates; its cost, the downward-biased effective beta required to hold the floor, was paid only on tickers where the unclipped identity could not be maintained.

We also asked whether the hyperparameters of Algorithm 1, the R^2-preserve threshold R^2_{preserve} and the idiosyncratic floor f, had been tuned to the headline result. We re-ran the hybrid composer on the full 423-ticker universe at every point of a 5×5 grid, $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$ and $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ (Fig. 6). Hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points, and the R^2-preserve assignment was stable at QQQ and SPYG for every threshold at or below $R^2 = 0.85$, with QQQ reassigning to the `hybrid` branch only at the $R^2_{\text{preserve}} = 0.90$ corner. The floor f did not change any branch assignment in the grid, since the maximum $\rho_i = \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ in the universe stayed below $1 - f$ for every $f \in \{0.05, \ldots, 0.30\}$. The headline number is therefore invariant to the hyperparameter values within wide tolerances, and the operating point ($R^2_{\text{preserve}} = 0.80$, $f = 0.10$) reflects a principled choice of the R^2-branch threshold and a standard floor value rather than a data-peeked local optimum.

Finally, we quantified the contribution of generator stochasticity to the headline numbers by re-running the full six-composer protocol under 8 random seeds (Table 4). The per-composer KS pass rates held to within fractions of a percentage point: hybrid sat at $50.40 \pm 0.18\%$ across seeds, JumpHMM-on-residuals at $75.61 \pm 0.11\%$, block bootstrap at $73.24 \pm 0.09\%$, GARCH(1,1)-t at $66.96 \pm 0.30\%$, naive at $4.41 \pm 0.07\%$, and Gaussian SIM at $0.60 \pm 0.04\%$. The across-composer differences therefore exceeded the seed-induced spread by roughly two orders of magnitude. The seed sweep also reproduced the relative ordering of the composers without inversions across seeds, confirming that the hybrid-vs-baseline gap and the residual-fit-vs-hybrid body-fidelity gap were not artifacts of the simulation seed.

4 Discussion

In this study, we composed synthetic per-asset growth-rate paths under a single-index model (SIM) by drawing each asset's innovation from an independent JumpHMM generator fit to the asset's own full growth-rate history, then rescaling the draw by a closed-form scalar before inserting it into the SIM composition. We evaluated six composers on a calibrated universe of 424 US equities and exchange-traded funds against the real SPY market path: the naive composition, a Gaussian-residual SIM baseline, the hybrid construction proposed here, and three residual-fit alternatives (JumpHMM-on-residuals, Politis–Romano block bootstrap [3], and a per-ticker GARCH(1,1)-t fit [2]). The one-line variance correction, together with the sign-preserving clip rule and the R^2-preserving branch, closes a gap between two previously unsatisfying options for building synthetic asset paths with both marginal structure and factor structure: naive composition gives both but distorts the marginal by $\beta^2\sigma_m^2$, while Gaussian-residual SIM gives the factor exactly but strips the marginal. The hybrid construction sat between the Gaussian-residual and naive baselines: it recovered β to the same precision as either, held the generator marginal to its target variance in every ticker, and preserved the generator's tails at a level indistinguishable from the naive composition. The three residual-fit composers produced higher Kolmogorov–Smirnov pass rates than hybrid on marginal body fidelity (67 to 76% versus 50%), but matched hybrid on the 99% Value-at-Risk coverage test to within statistical tolerance (Table 5); the body-fidelity gap did not translate to a tail-accuracy gap. Hybrid's contribution is therefore scoped: it is the best composition rule when the per-asset generator is specified in full-return units (a common setting, since per-asset generators are often calibrated for distributional interpretability, cached for other pipelines, or chosen to retain regime-switching or jump structure), and it delivers 99% VaR accuracy on par with a dedicated residual generator without requiring a separate fitting stage. The gap was large enough to matter in practice: at the high-β end of the universe, naive composition inflated the marginal variance by over 70% relative to the generator target (Fig. 1). Because branch selection is driven entirely by the calibrated R^2 and each asset carries a persisted construction flag, the procedure runs uniformly across an arbitrary ticker universe without external metadata and leaves an audit trail that downstream consumers can inspect. No part of the construction is new in a deep sense; the contribution is a minimal, practitioner-ready recipe with an auditable branch flag and a principled treatment of the degenerate tracker limit, usable as a drop-in replacement for naive composition in workflows where both tail shape and market dependence matter.

The evaluation isolated the composition rule by fixing the market path at the real SPY history and using the same per-ticker innovation draw across all three composers. This design yielded a paired comparison of composition rules at the per-ticker level but set aside two questions that production users will need to address separately. The first concerns cross-sectional dependence: we skipped the optional rank-reorder step in Algorithm 1 so that per-ticker metrics reflected the composition rule alone. When the reorder is applied, the variance algebra is unchanged (rank-reordering preserves the marginal variance of ε_i), so the marginal results here carry over; cross-sectional fidelity under different copula specifications is studied in the companion `JumpHMM.jl` paper [7]. The second concerns market-path uncertainty: the metrics reported here used the real SPY path, and under a simulated market path the composed marginal picks up additional variance from the sampling of $g_m^{(r)}$, which for high-R^2 tickers becomes a noticeable effect. In production, the two deferred questions compose: a reorder-plus-simulated-market pipeline would layer cross-sectional structure on top of per-ticker draws built against sampled market paths, with the variance correction of Eq. (4) still providing the per-asset variance target on each realized $g_m^{(r)}$. A separate scope boundary concerns the downstream use case: the main metrics reported here characterize marginal and factor fidelity of the composition rule on per-ticker targets. To test whether that fidelity translates to a consequential

downstream number, we also report a per-ticker one-day 95% and 99% Value-at-Risk backtest of the composed paths against the real 2014–2024 growth-rate histories, with exceedance counts assessed by Kupiec's unconditional coverage test; the per-ticker framing keeps this validation inside the paper's scope, while portfolio-level VaR requires the cross-sectional copula deferred to the companion paper [7]. High-turnover portfolio strategies remain a distinct scope boundary: the temporal structure of the residual draw becomes load-bearing in a way that contemporaneous composition does not address.

Clipping did not trigger anywhere in the US-equity universe used here, because the JumpHMM marginals carried enough variance that $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\mathrm{gen},i}^2$ stayed below 0.90 even for the highest-β ticker. Clipping becomes relevant in two settings we did not evaluate: scenario-driven markets, where the synthetic g_m is constructed to be much more volatile than the calibration market (e.g., stress tests), and universes that mix low-volatility single stocks with extreme betas (e.g., leveraged ETFs). In both settings we recommend reporting the original and effective β_i alongside the output, as in Algorithm 1, so that downstream consumers can audit the attenuation. A related attenuation appeared at the other end of the β spectrum, even when clipping did not trigger: the hybrid KS pass rate fell from 67% at low β to 44% at high β (Table 3), a mild attenuation that followed directly from the composition algebra. As $\beta_i^2 \sigma_m^2$ grows toward the idiosyncratic floor $(1-f)\sigma_{\mathrm{gen},i}^2$, the market-driven component carries an increasing share of the marginal, and that component is only as heavy-tailed as g_m. The market's own heavy tails slow the attenuation (SPY growth rates are themselves heavy-tailed after JumpHMM fitting), but the effect is visible. Users who need exact marginal fidelity at high β can recover it by lowering the floor f (at the cost of less conservative clipping behavior) or by pre-scaling the generator variance so that ρ_i stays small.

The variance correction $s_i^2 = 1 - \rho_i$ relies on $\mathrm{Cov}(\tilde{\varepsilon}_i, g_m) \approx 0$. The JumpHMM marginals are fit in isolation and sampled independently of the market path, so this holds by construction; if the generator is already correlated with the market (for example, a bootstrap from joint history), the empirical covariance should be subtracted from β_i before applying the correction, or s_i^2 should be estimated directly from the empirical variance of the composed series. The same warning applies to the R^2-preserving branch, whose variance identity assumes ε is uncorrelated with g_m after any reorder. The construction also injects a contemporaneous market factor only: $g_i(t)$ depends on $g_m(t)$ and on $\tilde{\varepsilon}_i(t)$ but not on $g_m(t-1), g_m(t-2), \ldots$, so lagged market dependence (lead-lag effects, market-driven volatility clustering, asymmetric responses to drawdowns) is not reproduced by this recipe. The natural extension is a lagged-factor SIM, $g_i(t) = \alpha_i + \sum_{k=0}^{K} \beta_{i,k}\, g_m(t-k) + \varepsilon_i(t)$; the variance correction generalizes to that form by replacing $\beta_i^2 \sigma_m^2$ with the variance of the full market-factor component, though the empirical value of the extra parameters is beyond the scope here. Lagged idiosyncratic dependence is likewise absent: the JumpHMM residuals $\tilde{\varepsilon}_i$ are drawn independently across time, so any residual autocorrelation present in the real data is discarded by the composition. This matters most for workflows that evaluate portfolios rebalancing at high frequency. With iid residuals, a rebalancing rule collects the full diversification return of [8], a compounding premium that scales with residual variance and rebalancing frequency and is largely absent in real markets, where daily residual autocorrelation is small but positive [9]. Two generator alternatives in Table 1 preserve temporal structure directly: the block-bootstrap composer resamples blocks of the empirical OLS residual series [3], inheriting whatever autocorrelation is present in the data, and the GARCH(1,1)-t composer fits a conditional-variance model per ticker [2]. Both recover marginal fidelity at the cost of committing the pipeline to a specific residual generator, whereas the variance correction of Eq. (4) is generator-agnostic and accepts a full-return JumpHMM marginal (with its jump and regime structure) as input. Users evaluating high-turnover strategies on JumpHMM-corrected paths should benchmark absolute return levels against a static-allocation baseline on the same paths, or adopt the bootstrap or GARCH alternatives when residual autocorrelation is the load-bearing structure for the downstream task. Two further extensions follow naturally. Multi-factor SIM replaces the single market factor with K factors, requiring ρ_i to become the sum of factor contributions and the loading vector to be clipped rather than a scalar, though the branch logic is unchanged. Per-ticker tuning of the JumpHMM jump parameters (ϵ, λ) tightens each marginal's fit; we used defaults to keep fitting tractable across the universe, but tuning as reported in the companion paper [7] would push the hybrid KS pass rates upward. A stress-test harness ties these together: using the clipping rule to enforce a minimum idiosyncratic share in constructed market scenarios guarantees that per-asset marginals remain heavy-tailed even under synthetic markets much more volatile than history, which is precisely the regime where naive SIM composition breaks down most severely.

Acknowledgments

The author thanks Abdulrahman Alswaidan and the eCornell AI in Finance program participants for discussions that motivated the variance-correction formulation, and Polygon.io for access to the 2014 to 2024 US equity and ETF daily price histories used in the experiments. Computations were run on a single laptop; no shared-resource allocation was required.

Author contributions. J.D.V. developed the method, wrote the code, performed the experiments, and wrote the paper.

Competing interests. The author declares no competing interests.

Code and data availability. Source code and experiment scripts are available at https://github.com/varnerlab/hybrid-sim-composition. The per-asset hidden Markov model generators rely on `JumpHMM.jl` (https://github.com/varnerlab/JumpHMM.jl).

References

[1] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.

[2] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.

[3] Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.

[4] William F. Sharpe. A simplified model for portfolio analysis. *Management Science*, 9(2):277–293, 1963. doi: 10.1287/mnsc.9.2.277.

[5] Stephen A. Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13 (3):341–360, 1976. doi: 10.1016/0022-0531(76)90046-6.

[6] Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993. doi: 10.1016/0304-405X(93)90023-5.

[7] Abdulrahman Alswaidan and Jeffrey D. Varner. JumpHMM.jl: Heavy-tailed per-asset marginal models for synthetic return generation. arXiv preprint, March 2026, 2026. companion paper.

[8] David G. Booth and Eugene F. Fama. Diversification returns and asset contributions. *Financial Analysts Journal*, 48(3):26–32, 1992. doi: 10.2469/faj.v48.n3.26.

[9] Paul Bouchey, Duncan Gunthorp, and Matt Sutherland. Volatility harvesting in theory and practice. *Journal of Wealth Management*, 15(2):89–100, 2012.

Table 1: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composer. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. All three composers recover β to the same precision; only the hybrid composer preserves the marginal distribution (KS pass 50.4%) while holding the variance close to the generator target.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

Table 2: **Hybrid composer broken out by construction flag.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The `hybrid-clipped` branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the `r2-preserve` branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

Table 3: **Aggregate metrics by** β **quartile, across composers.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat of Section 2.6; naive collapses from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table 4: **Seed-uncertainty summary across** 8 **random seeds.** Each cell reports mean $\pm$ SD across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker $\times$ 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The between-composer differences in KS pass rate exceed the seed-induced SD by roughly two orders of magnitude (hybrid SD 0.18 pp vs 25 pp gap to JumpHMM-on-residuals).

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000

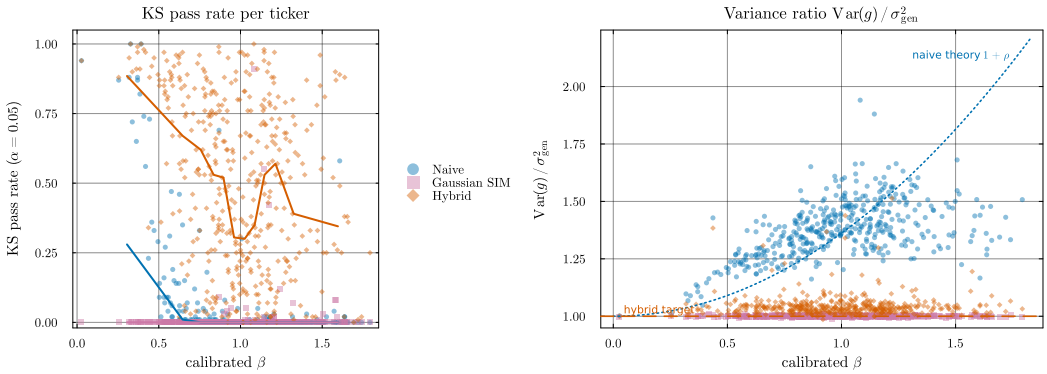

Figure 1: **Marginal-distribution preservation and variance preservation vs calibrated** β**.** Each point is a per-ticker median over 100 replications. *Left:* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$. The hybrid composer (orange) maintains a pass rate between ~ 0.3 and ~ 0.9 across the full β range; the naive composer (blue) collapses past $\beta \approx 0.7$ as the market loading inflates the marginal variance; the Gaussian SIM baseline (purple) is pinned at zero throughout. *Right:* variance ratio $\mathrm{Var}(g)/\sigma^2_{\text{gen}}$. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the naive scatter follows it. Hybrid and Gaussian SIM sit on the unit line by design.

Table 5: **Per-ticker one-day Value-at-Risk backtest across the six composers.** For each (ticker, replication) pair we estimate the VaR threshold at coverage level α from the composed path and count exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1-\alpha$). *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable at both coverage levels; both strawman composers (naive, Gaussian SIM) fail coverage on a majority of the universe.

	$\alpha = 0.95$		$\alpha = 0.99$	
Composer	rate (%)	Kupiec pass (%)	rate (%)	Kupiec pass (%)
Naive	3.55	10.4	0.67	45.2
Gaussian SIM	4.06	42.6	1.40	47.0
Hybrid	4.75	80.4	0.95	79.7
JumpHMM-on-residuals	4.87	84.0	0.99	82.9
Block bootstrap	4.78	76.7	1.09	86.0
GARCH(1,1)-t	4.64	69.0	1.09	72.1

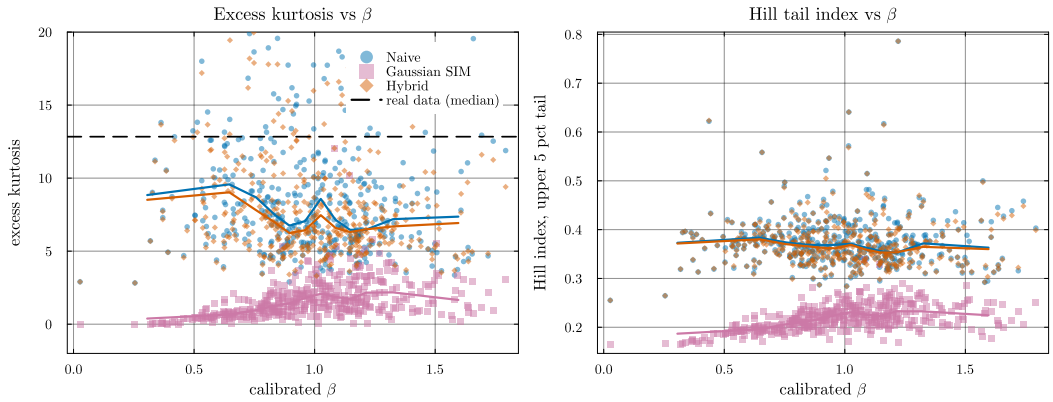

Figure 2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composers (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians sit near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median sitting below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion; the panel's point is that hybrid tracks naive (inheriting generator tails) while Gaussian SIM does not.

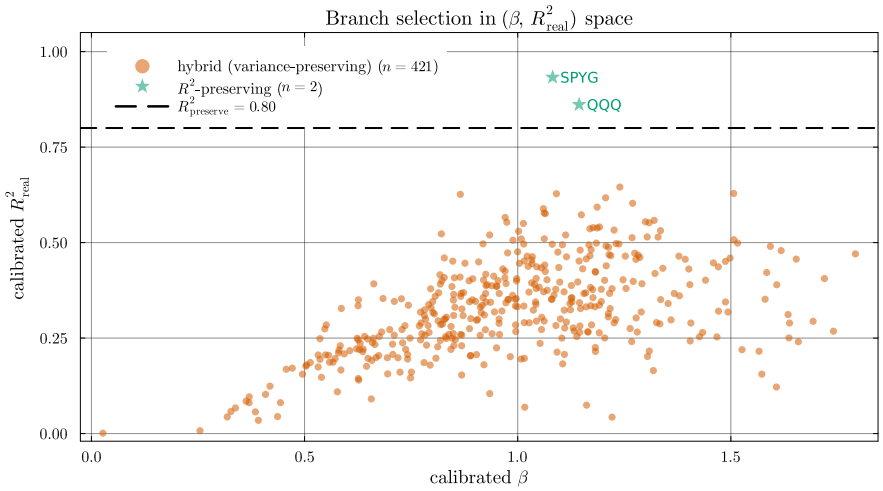

Figure 3: **Construction-flag assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from genuine single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the `r2-preserve` branch. The remaining 421 tickers populate the `hybrid` (variance-preserving) branch; the `hybrid-clipped` branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

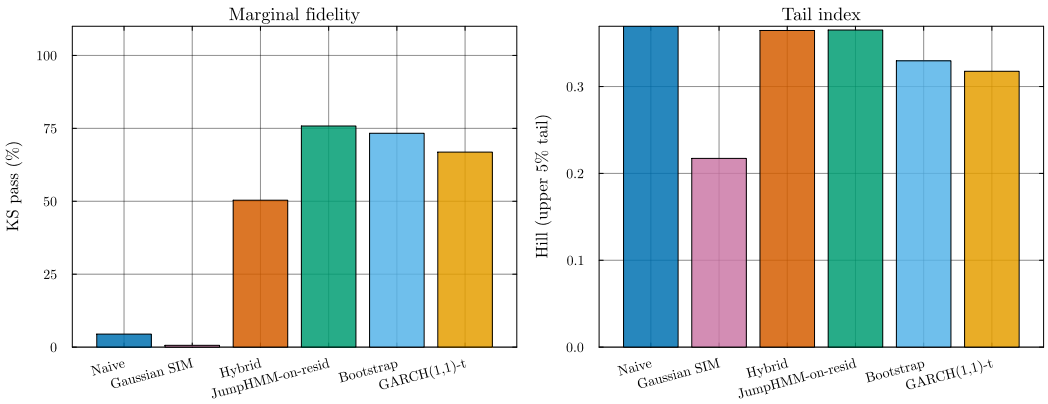

Figure 4: **Six-composer comparison on aggregate marginal fidelity and tail index.** *Left:* per-ticker Kolmogorov–Smirnov pass rate at $\alpha = 0.05$, aggregated across 423 tickers and 100 replications per ticker. The three residual-fit composers (JumpHMM-on-residuals, block bootstrap, GARCH(1,1)-t) dominate the hybrid on marginal body fidelity; both strawman composers collapse. *Right:* median Hill tail index on the upper 5% of $|g|$. Hybrid, JumpHMM-on-residuals, and naive cluster at ~ 0.37, reflecting the shared full-return JumpHMM generator's tail character; block bootstrap and GARCH sit slightly lower at 0.33–0.32; Gaussian SIM collapses to 0.22.

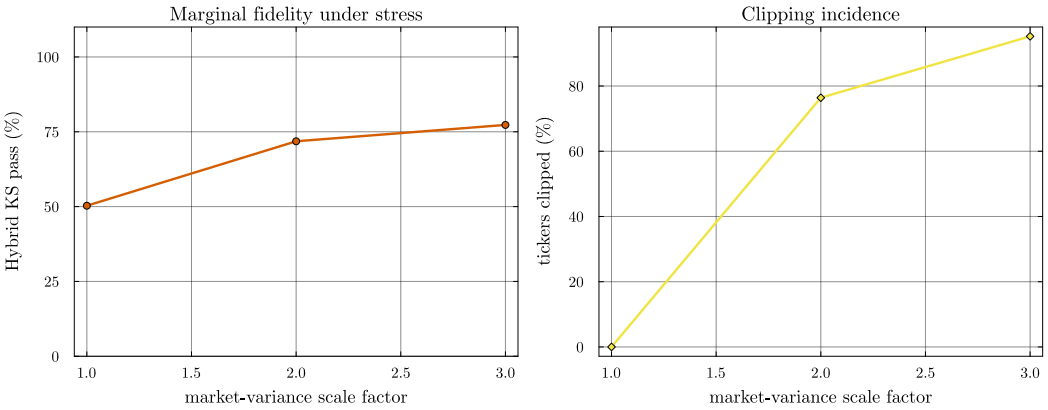

Figure 5: **Clipping-branch behavior under market-variance stress.** Hybrid composer re-run with the SPY growth-rate path rescaled by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed. *Left:* hybrid KS pass rate at $\alpha = 0.05$ rises from the baseline 50.3% at $\gamma = 1$ to 71.8% and 77.3% at $\gamma = 2$ and $\gamma = 3$, because the clipping branch rescales β^{eff} to hold the idiosyncratic floor and prevents the market component from crowding out the generator's marginal. *Right:* fraction of tickers for which the clipping branch activated ($\rho_i > 1 - f = 0.90$). No ticker triggered clipping in the main run; doubling σ_m pushed 76% of the universe past the threshold, and $3\times$ pushed 95%. The branch is a robustness mechanism, not a degradation mode.

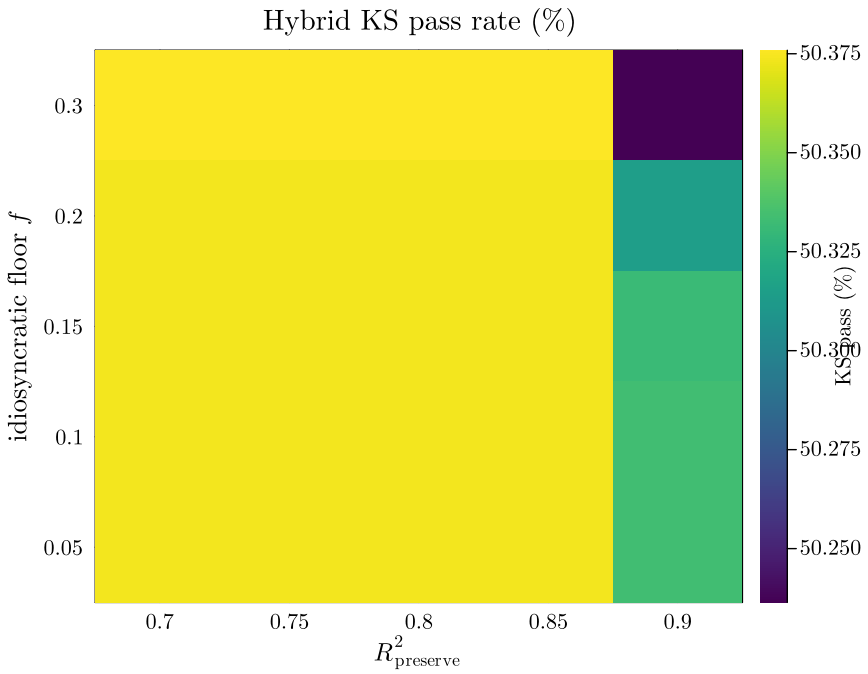

Figure 6: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The operating point $(0.80, 0.10)$ used throughout the paper is indistinguishable from every other grid point within sampling noise, indicating that the headline result is not a data-peeked local optimum.

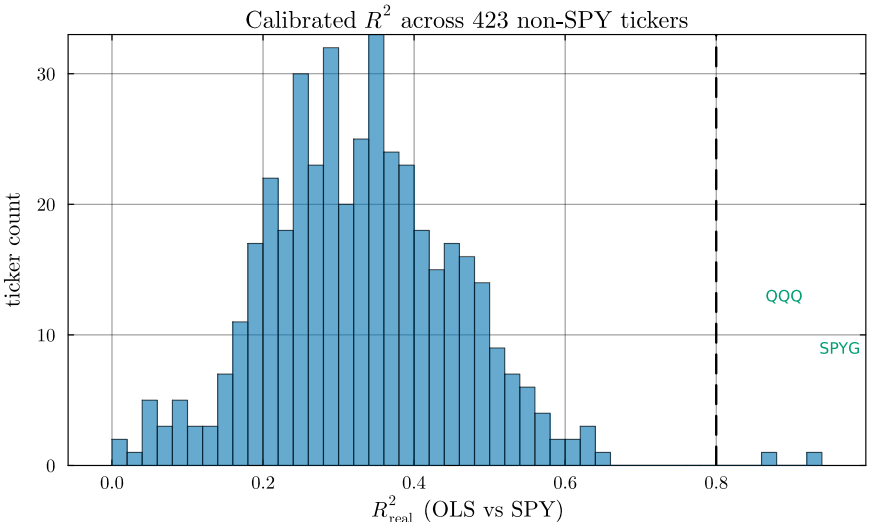

Figure 7: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) sits a structural ~ 0.22 below the threshold. The `r2-preserve` branch's $n = 2$ occupancy is therefore a property of the single-factor SIM on S&P 500 names against SPY, not a sample-size artifact or a data-peeked threshold.

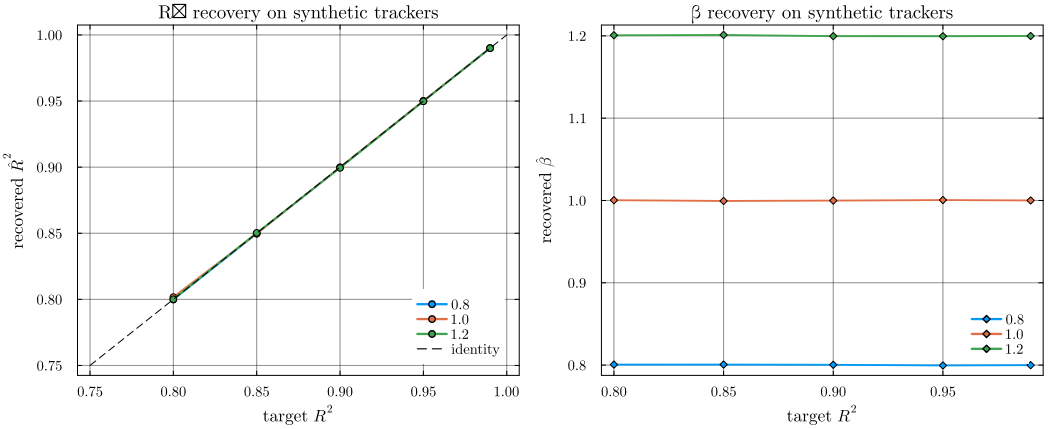

Figure 8: R^2**-preserve branch validated on synthetic trackers.** For each $(\beta_{\text{true}}, R^2_{\text{true}})$ with $\beta_{\text{true}} \in \{0.8, 1.0, 1.2\}$ and $R^2_{\text{true}} \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$, a synthetic tracker was constructed from $\sigma^2_{\varepsilon,\text{true}} = \beta^2_{\text{true}}\sigma^2_m(1 - R^2_{\text{true}})/R^2_{\text{true}}$ and composed via Algorithm 1 at 100 replications. *Left:* recovered $\hat{R}^2$ against target R^2, grouped by β_{true}; the identity line is dashed. *Right:* recovered $\hat{\beta}$ against target R^2. Across all 15 cells the branch recovered median $\hat{\beta}$ to within 0.001 of β_{true}, median $\hat{R}^2$ to within 0.001 of R^2_{true}, and cleared the $\alpha = 0.05$ KS test on at least 99% of replications per cell.

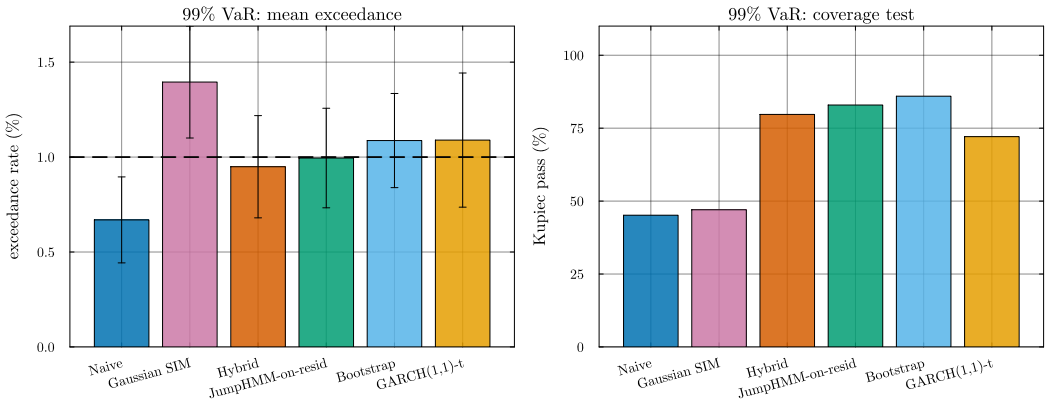

Figure 9: 99% **Value-at-Risk exceedance rates and Kupiec coverage across the six composers.** *Left:* mean exceedance rate across tickers (nominal 1%, dashed line); error bars are one SD across tickers. Hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t all hit coverage within 0.1pp of nominal; naive under-breaches (inflated variance in the composed marginal) and Gaussian over-breaches (thin tails). *Right:* fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable (80 and 83%); the strawman composers fail on the majority of the universe.

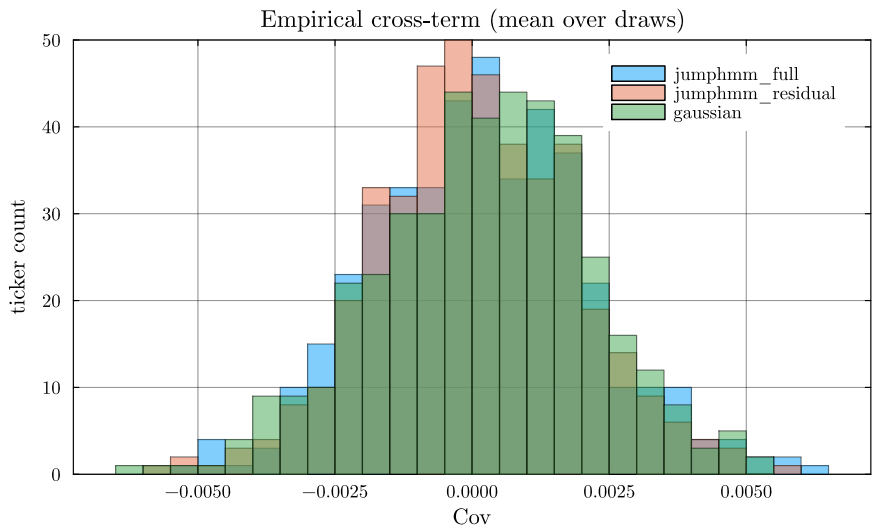

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m]/(\sigma_{\tilde{\varepsilon},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms are stacked per generator, with each per-ticker value averaged over $R = 100$ innovation draws. The full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators all cluster at zero, validating the independence assumption used to derive Eq. (4) in Section 2.3.

A Empirical cross-term diagnostic

The variance-correction in Section 2.3 rests on the assumption $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$, which is what allows the derivation of Eq. (4) to drop the cross-term in the expansion of $\mathrm{Var}[g_i]$. Because the per-asset JumpHMM marginals are fit to each asset's full growth-rate history, including its market-correlated component, the assumption is not automatic: a generator that picked up a residual market loading would invalidate the closed-form scaling. We checked the assumption empirically by sampling 100 innovation paths $\tilde{\varepsilon}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R} \frac{\mathrm{Cov}[\tilde{\varepsilon}_i^{(r)},\, g_m]}{\sigma_{\tilde{\varepsilon},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values are plotted as histograms in Fig. S1. The full-return JumpHMM generator (`jumphmm_full`, used by hybrid and naive) sits within ± 0.05 of zero with a slight rightward bias of order 10^{-2}; the residual-fit JumpHMM generator (`jumphmm_residual`, used by JumpHMM-on-residuals) and the Gaussian baseline are both centered at zero with no detectable bias. The independence assumption is therefore satisfied to within sampling tolerance for every composer that uses it, and the derivation of Eq. (4) holds in practice.